

Homework Template

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Problem 1

Given $n = 11413 = 101 \cdot 113$ and $e = 7467$. Then

$$\varphi(n) = (101 - 1)(113 - 1) = 11200.$$

Find $d \equiv e^{-1} \pmod{\varphi(n)}$. Since

$$7467 - 2(11200 - 7467) = 3 \cdot 7467 - 2 \cdot 11200 = 1,$$

we have $d \equiv 3 \pmod{11200}$, so $d = 3$. Decrypt:

$$m \equiv 5859^3 \pmod{11413}.$$

Compute $5859^2 \equiv 8990 \pmod{11413}$ and $8990 \cdot 5859 \equiv \boxed{1415} \pmod{11413}$. Thus the plaintext is $\boxed{1415}$.

Problem 2

$p = 23$, $q = 19$, so $n = pq = 437$ and $\varphi(n) = (23 - 1)(19 - 1) = 396$. With $e = 397 \equiv 1 \pmod{396}$,

$$d \equiv e^{-1} \equiv 1 \pmod{396}.$$

Hence encryption and decryption are the identity, so the plaintext is

$$\boxed{m \equiv c = 123 \pmod{437}}.$$

Problem 3

$n = 979 = 11 \cdot 89$, so $\varphi(n) = (11 - 1)(89 - 1) = 880$. With $e = 587$,

$$1 = 587 - 2(880 - 587) = 3 \cdot 587 - 2 \cdot 880 \Rightarrow d \equiv 3 \pmod{880}.$$

Decrypt $c = 10$:

$$m \equiv 10^3 = 1000 \equiv \boxed{21} \pmod{979}.$$

Problem 4

$n = 667$, $e = 3$, $c = 62$. The plaintext is either 9 or 10. Compute $9^3 = 729 \equiv 62 \pmod{667}$, while $10^3 = 1000 \not\equiv 62 \pmod{667}$. Therefore the plaintext is $\boxed{9}$.

Problem 5

Here $n = p = 1093$ is prime, $e = 361$. Then $\varphi(n) = n - 1 = 1092$ and Eve should take

$$d \equiv e^{-1} \pmod{1092}.$$

Euclid: $1092 = 3 \cdot 361 + 9$, $361 = 40 \cdot 9 + 1$, so

$$1 = 361 - 40(1092 - 3 \cdot 361) = 121 \cdot 361 - 40 \cdot 1092.$$

Hence $d \equiv \boxed{121} \pmod{1092}$, and Eve decrypts by $m \equiv c^{121} \pmod{1093}$.

Problem 6

$e = 1$ is invalid because encryption is $c \equiv m^1 \equiv m \pmod{n}$ (no secrecy). $e = 2$ should not be used because for RSA with $n = pq$ (odd primes) we have $\varphi(n) = (p-1)(q-1)$ even (indeed a multiple of 4), so $\gcd(2, \varphi(n)) \neq 1$ and $d = e^{-1} \pmod{\varphi(n)}$ does not exist. Even if it did in some contrived case, such a tiny exponent would be insecure.

Problem 7

Alice encodes each letter $a = 1, \dots, z = 26$ separately and sends $m^e \pmod{n}$ with $n = 8881$, $e = 13$. Since the message space is only $\{1, \dots, 26\}$, Eve can build a lookup table of $m^{13} \pmod{8881}$ for $m = 1, \dots, 26$ and match each ciphertext block.

Applying this table to the intercepted sequence

$$4461, 794, 2015, 2015, 3603$$

yields the letters

$$\boxed{\text{H E L L O}}.$$

Thus the message is $\boxed{\text{HELLO}}$ (no factoring of n needed; just a 26-entry lookup).

Problem 8

Suppose $n = pqr$ is the product of three distinct primes. How would an RSA-type scheme work in this case? What relation would e and d satisfy?

Solution.

If $n = pqr$ with distinct primes p, q, r , then an RSA-type system works the same way as usual, but the Euler totient becomes

$$\varphi(n) = (p-1)(q-1)(r-1).$$

To choose an encryption exponent e , we require

$$\gcd(e, \varphi(n)) = 1.$$

Then the decryption exponent d must satisfy

$$ed \equiv 1 \pmod{\varphi(n)}.$$

Encryption and decryption work as usual:

$$c \equiv m^e \pmod{n}, \quad m \equiv c^d \pmod{n}.$$

The only change from standard RSA (two primes) is the definition of $\varphi(n)$. Otherwise, the construction and correctness of the scheme remain the same.

Problem 9

A trusted entity chooses primes p, q , sets $n = pq$, and chooses integers k_1, k_2, k_3 satisfying

$$k_1 k_2 k_3 \equiv 1 \pmod{(p-1)(q-1)}.$$

Keys:

$$\text{Alice: } (n, k_1, k_2), \quad \text{Bob: } (n, k_2, k_3), \quad \text{Carla: } (n, k_1, k_3).$$

(a)

Alice wants Bob and Carla to decrypt the same message m . She encrypts using the key component they *both* possess: Bob has k_2 , Carla also has k_2 ? No—Carla has k_1, k_3 . Bob has k_2, k_3 .

The common exponent between Bob and Carla is k_3 .

Thus Alice sends:

$$c \equiv m^{k_3} \pmod{n}.$$

Bob decrypts using k_2 since $k_2 k_3 k_1 \equiv 1$. Carla decrypts using k_1 .

(b)

Alice wants only Carla to decrypt. She should use an exponent that: - Carla knows, - Bob does *not* know.

Carla knows k_1 and k_3 ; Bob knows k_2 and k_3 . So the exponent Bob does not know is k_1 .

Alice sends:

$$c \equiv m^{k_1} \pmod{n}.$$

Carla decrypts using $k_2 k_3$, but Bob cannot decrypt.

Problem 10

Secret: 5. Four people A, B, C, D . Any two of them must reconstruct the secret; no one alone can.

Use a $(2, 4)$ Shamir scheme over a prime modulus (choose a prime e.g. $p = 17$). Let the secret be $f(0) = 5$ and choose a random slope $a \in \{1, \dots, 16\}$. Example: take $a = 7$.

Polynomial:

$$f(x) = 7x + 5 \pmod{17}.$$

Give each person one point:

$$A : (1, f(1) = 12), \quad B : (2, f(2) = 19 \equiv 2), \quad C : (3, f(3) = 26 \equiv 9), \quad D : (4, f(4) = 33 \equiv 16).$$

Any two points determine the line and recover $f(0) = 5$. No single point reveals the secret.

Problem 11

A $(2, 10)$ Shamir scheme over prime 73. Two known shares:

$$(1, 10), \quad (2, 18).$$

Polynomial is linear: $f(x) = ax + b \pmod{73}$. From $f(1) = 10$ we get $a + b \equiv 10$. From $f(2) = 18$ we get $2a + b \equiv 18$.

Subtract to find $a \equiv 8$. Then $b \equiv 10 - 8 = 2$.

Thus

$$f(x) = 8x + 2 \pmod{73}.$$

The third share at $x = 5$ is:

$$f(5) = 8 \cdot 5 + 2 = 42.$$

So the missing value is

$$\boxed{42}.$$

Problem 12

A $(3, 5)$ Shamir scheme over $p = 17$. Shares:

$$(1, 8), (3, 10), (5, 11).$$

Use Lagrange interpolation to compute $f(0)$.

$$f(0) = 8 \cdot \ell_1(0) + 10 \cdot \ell_3(0) + 11 \cdot \ell_5(0) \pmod{17}.$$

Compute basis terms:

$$\ell_1(0) = \frac{(0-3)(0-5)}{(1-3)(1-5)} = \frac{(-3)(-5)}{(-2)(-4)} = \frac{15}{8} \equiv 15 \cdot 8^{-1} \equiv 15 \cdot 15 \equiv 4,$$

$$\ell_3(0) = \frac{(0-1)(0-5)}{(3-1)(3-5)} = \frac{(-1)(-5)}{(2)(-2)} = \frac{5}{-4} \equiv 5 \cdot (-4)^{-1} = 5 \cdot (-4)^{-1}.$$

Since $(-4) \equiv 13$, $13^{-1} \equiv 4$. Thus $\ell_3(0) = 5 \cdot 4 = 20 \equiv 3$.

$$\ell_5(0) = \frac{(0-1)(0-3)}{(5-1)(5-3)} = \frac{(-1)(-3)}{(4)(2)} = \frac{3}{8} \equiv 3 \cdot 8^{-1} \equiv 3 \cdot 15 = 45 \equiv 11.$$

Now compute:

$$f(0) \equiv 8 \cdot 4 + 10 \cdot 3 + 11 \cdot 11 = 32 + 30 + 121 = 183 \equiv \boxed{13} \pmod{17}.$$

So the secret is $\boxed{13}$.

Problem 13

We need a Shamir scheme where: - The general alone can launch, - Or the two colonels together, - Or all five clerks together, - But no smaller group can.

Use a $(1, 1)$, $(2, 2)$, and $(5, 5)$ structure combined under a higher global threshold.

Construct a $(1, 10)$ Shamir scheme: - Give the general share $(0, s)$ directly (he can recover alone). - For the two colonels, embed the secret into a $(2, 2)$ subsystem, distribute both shares among them. - For the clerks, embed into a $(5, 5)$ subsystem, distribute the five shares.

All these subschemes encode the same secret. Any authorized group gets all shares of at least one full subsystem; any unauthorized group does not.

Thus the launch conditions are satisfied.

Problem 14

We want the launch code to be reconstructible only if

$$3 \text{ from A, } 4 \text{ from B, } 2 \text{ from C}$$

cooperate. Use a two-level Shamir construction.

Level 1: Between the three delegations.

Let the launch code be the secret S , working modulo a large prime p . Choose a random quadratic polynomial

$$f(x) = ax^2 + bx + S \pmod{p},$$

so this is a $(3, 3)$ -threshold scheme.

Give the three delegations the “group shares”

$$S_A = f(1), \quad S_B = f(2), \quad S_C = f(3).$$

Any three of these (i.e., all of them) are needed to reconstruct S .

Level 2: Inside each delegation.

Now treat S_A, S_B, S_C as secrets and share them within each group:

- For delegation A, use a $(3, 10)$ Shamir scheme with secret S_A . Each of the 10 members of A receives one share. Any 3 of them can recover S_A , but 2 or fewer cannot.
- For delegation B, use a $(4, 10)$ Shamir scheme with secret S_B . Any 4 members of B together can reconstruct S_B .
- For delegation C, use a $(2, 10)$ Shamir scheme with secret S_C . Any 2 members of C together can reconstruct S_C .

Reconstruction.

To reconstruct the launch code:

1. At least 3 members of A combine their shares to obtain S_A .
2. At least 4 members of B combine theirs to obtain S_B .
3. At least 2 members of C combine theirs to obtain S_C .
4. Using the three points $(1, S_A), (2, S_B), (3, S_C)$ on $f(x)$, they interpolate f and compute $f(0) = S$, the launch code.

Any set that is missing at least one of these group secrets cannot obtain all three points on $f(x)$, so they cannot reconstruct the launch code.

References